


# Business Analytics

2024-10-01 to 2026-09-30 · Demo Client

| | |
|---|---|
| **Prepared for** | Demo Client |
| **Engagement** | DEMO-0011 |
| **Report date** | 11 November 2026 |
| **Version** | 1.0 |
| **Prepared by** | ProQuant Analytics · info@proquantanalytics.com |
| **Run ID** | ba_20261111T142253Z_f9025f |

# Executive summary

**Demonstration report.** Built entirely from synthetic data generated by ProQuant for illustration: a café-restaurant chain that does not exist. No real business, customer or person is described.

ProQuant Analytics brought together every record Demo Client keeps (expenses, items, marketing, customers, places, staff, targets, transactions) for 2024-10-01 to 2026-09-30, and asked where the business makes and loses money, what moves its sales, and what to do next. The data came in the client's own formats and was mapped to one standard set of tables; what was left out, and why, is listed in section 11.

**Overall view:** The business needs attention

7 of 17 health tests passed, 6 with a warning, 4 failed. The thresholds were agreed before the results were seen.

SALES, LAST 12 MONTHS
**SAR 4.98m**
+23.7% recently, against the same months a year earlier

GROSS MARGIN
**60.6%**
sales less the cost of what was sold

OPERATING PROFIT
**SAR -395k**
-7.9% of sales, after head office

ORDERS, 12 MONTHS
**67,022**
3 branches

NEW GUESTS WHO RETURN
**89%**
within 90 days

NEXT SIX MONTHS
**SAR 3.19m**
range 2.40m to 4.24m

## Findings

- **Sales** Sales were SAR 4.98m over the last 12 months; the recent months are +23.7% against the same months a year earlier.
- **Profit** After every cost, including head office, the business lost SAR 395k (-7.9% of sales).
- **Branches** Loss-making branches: Airport, with 36% of sales.
- **Dishes** Popular but low margin: Family Platter.
- **Guests** 89% of new guests come back within 90 days; a new one brings SAR 1,115 in a year.
- **Prices** After the +16% price change on Mains in 2025-10, units moved -7%.
- **Marketing** Snapchat took 43% of the spend and returned 0.68 of gross profit per unit spent.
- **Outlook** Next six months: about SAR 3.19m of sales (range SAR 2.40m to SAR 4.24m).
- **Analyst** Illustrative note (synthetic data): the Airport lease runs to 2027 with a break clause in March; the owner asked for the cost of leaving early.

## Actions

| Ref | Priority | Action | Owner | Due |
|---|---|---|---|---|
| A-01 | High | Win back the customers lost since the price rise; review the rise on the items that moved most. | Marketing | 2026-11-04 |
| A-02 | High | Reprice or redesign the popular low-margin items. | Operations manager | 2026-11-04 |
| A-03 | High | Agree a profit plan: start with the loss-making sites and the low-margin items. | Owner | 2026-11-04 |
| A-04 | High | Decide on each loss-making site: renegotiate the rent, cut costs, or close. | Owner | 2026-11-04 |
| A-05 | Medium | Recover the missing days from the system or confirm the site was closed. | Data owner | 2026-12-04 |
| A-06 | Medium | Add the missing items and their costs to the item list. | Finance | 2026-12-04 |
| A-07 | Medium | Move spend from the weakest channel to the best one; review in a month. | Marketing | 2026-12-04 |
| A-08 | Medium | Match staff hours to the busy hours at the weakest site. | Operations manager | 2026-12-04 |
| A-09 | Medium | Reset the targets or agree a plan with the site managers who miss them. | Finance | 2026-12-04 |
| A-10 | Medium | Use the forecast range, not the single number, until the error falls. | Finance | 2026-12-04 |

Source: tables/actions.csv

# Methodology

This report tells what was found and what it means for the business. How each number is built is in the methodology document; the definition of every measure is in the measure catalogue. One thing helps when reading on: profit by branch counts every cost booked to that branch, while head-office costs are shown on their own and never spread over the branches.

# Results

## 1. Sales and the months ahead

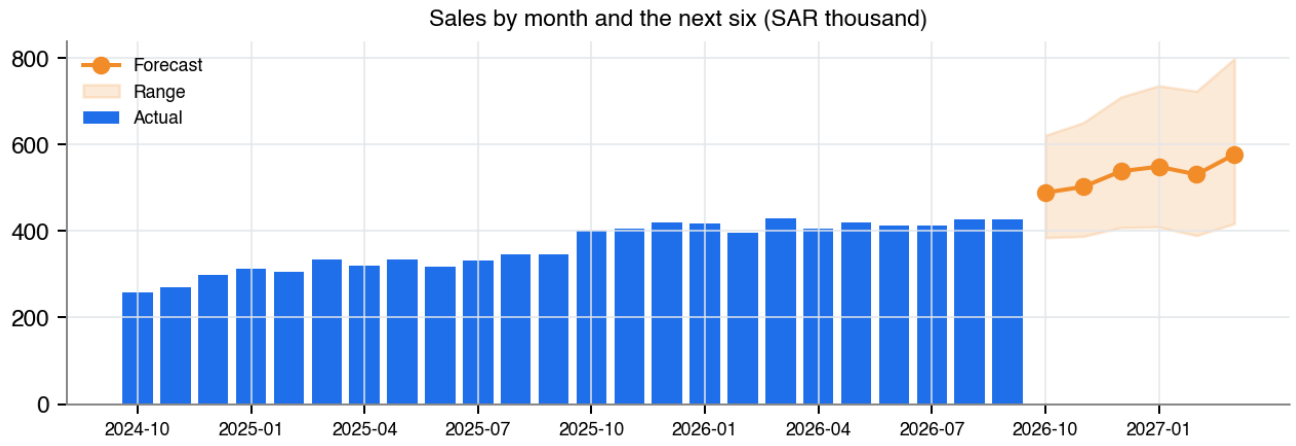



Figure 1. Sales by month, with the forecast and its range (trend and season). Source: tables/forecast.csv

Sales over the last twelve months were SAR 4.98m. The latest three months are +23.7% against the same months a year earlier. Held back and forecast, the last 6 months were missed by 10.9% on average, which is why the range matters more than the single line.

## 2. What moved sales

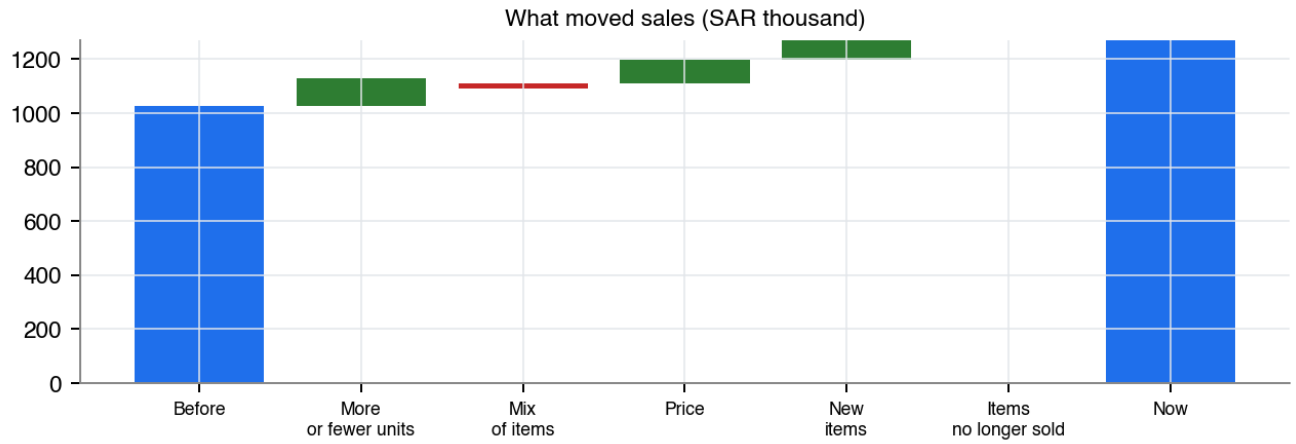



Figure 2. The change in sales, 2026-07 to 2026-09 against the same months a year earlier. Source: tables/drivers.csv

| Step | SAR |
|---|---|
| Before | 1,025,615 |
| More or fewer units | 102,272 |
| Mix of items | -17,547 |
| Price | 85,688 |
| New items | 72,283 |
| Items no longer sold | 0 |
| Now | 1,268,311 |

Source: tables/drivers.csv

The parts add up to the change exactly: more or fewer units at the old prices, a shift towards dearer or cheaper dishes, price, and dishes that came or went. The dishes that moved most:

| Dish | Before | Now | Change | Price |
|---|---|---|---|---|
| Water | 9,976 | 11,797 | 1,821 | -0.1% |
| Garden Salad | 39,240 | 42,496 | 3,256 | -0.0% |
| Soft Drink | 16,986 | 20,388 | 3,402 | +0.0% |
| Hummus | 24,795 | 28,508 | 3,713 | +0.0% |
| Grilled Chicken | 132,251 | 156,666 | 24,415 | +15.9% |
| Beef Burger | 129,091 | 156,402 | 27,310 | +14.7% |
| Family Platter | 165,699 | 199,538 | 33,839 | +0.1% |
| M06 (not in the item list) | 0 | 72,283 | 72,283 | n/a |

Source: tables/drivers_items.csv

## 3. Profit by branch

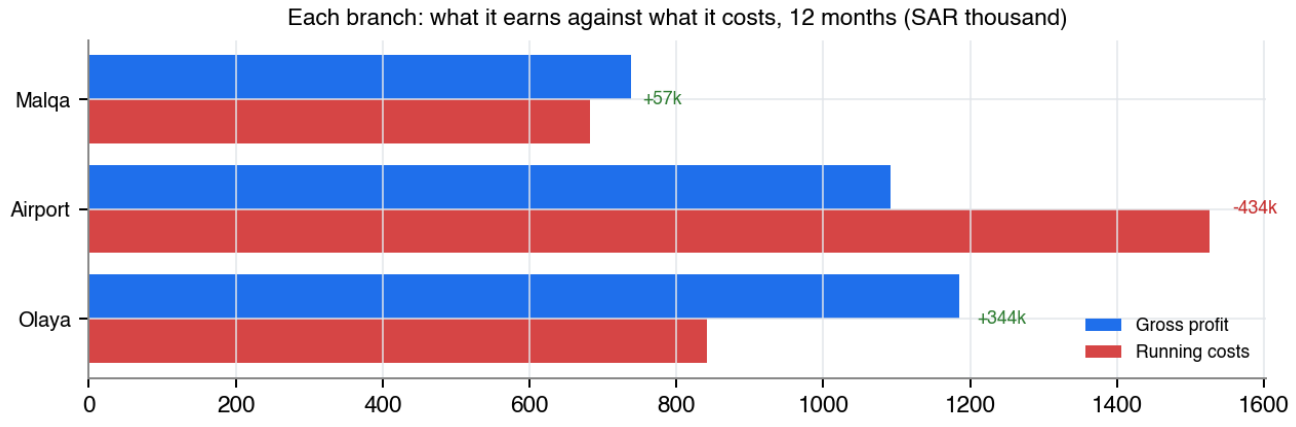



Figure 3. Gross profit against running costs, last 12 months. Source: tables/place_pnl.csv

| Branch | Sales | Cost of sales | Gross profit | Delivery commissions | Head office | Maintenance | Marketing | Other operating | Rent | Salaries | Utilities | Opera profit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Olaya | 1,966,331 | 780,612 | 1,185,719 | 90,238 | 0 | 25,188 | 0 | 36,159 | 240,000 | 375,941 | 73,736 | 344 |
| Airport | 1,795,188 | 703,457 | 1,091,731 | 84,597 | 0 | 27,101 | 0 | 37,227 | 900,000 | 406,421 | 70,501 | -434 |
| Malqa | 1,218,130 | 479,319 | 738,811 | 58,023 | 0 | 28,974 | 0 | 39,244 | 144,000 | 339,783 | 71,728 | 57 |

Source: tables/place_pnl.csv

Head-office costs of SAR 362k come on top and belong to no single branch.

Compared fairly, by size:

| Branch | Orders | Sales per seat a month | Sales per m² a month | Sales per staff hour | Share of sales |
|---|---|---|---|---|---|
| Olaya | 26,237 | 2,048 | 546 | 94 | 39% |
| Airport | 24,266 | 2,992 | 831 | 81 | 36% |
| Malqa | 16,519 | 1,692 | 461 | 66 | 24% |

Source: tables/place_pnl.csv

## 4. Dishes: what sells and what earns

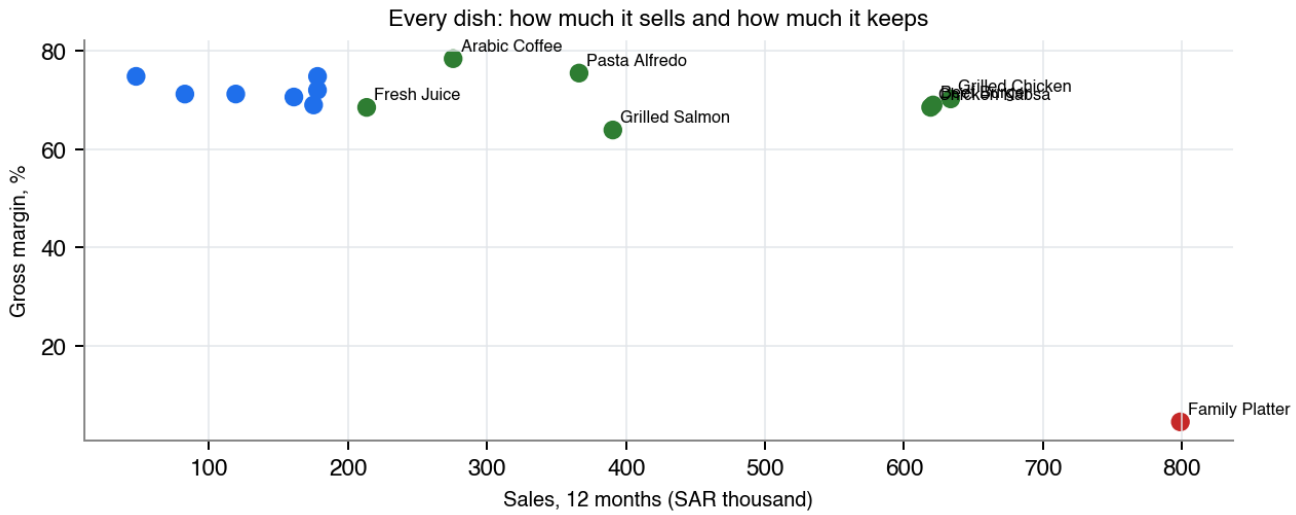



Figure 4. Sales against gross margin; red marks a popular dish with a low margin. Source: tables/items.csv

| Dish | Category | Sales | Units | Margin | Share | Group |
|---|---|---|---|---|---|---|
| Family Platter | Sharing | 798,808 | 8,476 | 4.5% | 16.0% | Popular, low margin |
| Grilled Chicken | Mains | 633,574 | 14,518 | 70.2% | 12.7% | Strong |
| Beef Burger | Mains | 620,896 | 16,053 | 69.0% | 12.5% | Strong |
| Chicken Kabsa | Mains | 619,123 | 13,003 | 68.5% | 12.4% | Strong |
| Grilled Salmon | Mains | 390,503 | 5,876 | 63.9% | 7.8% | Strong |
| Pasta Alfredo | Mains | 366,174 | 9,982 | 75.5% | 7.4% | Strong |
| Arabic Coffee | Drinks | 275,594 | 19,848 | 78.4% | 5.5% | Strong |
| Fresh Juice | Drinks | 213,374 | 13,449 | 68.5% | 4.3% | Strong |
| Fries | Sides | 178,021 | 14,960 | 74.8% | 3.6% | Profitable, slow |
| Garden Salad | Sides | 177,988 | 9,967 | 72.0% | 3.6% | Profitable, slow |
| Kunafa | Desserts | 175,196 | 6,793 | 69.0% | 3.5% | Profitable, slow |
| Cheesecake | Desserts | 161,009 | 6,763 | 70.6% | 3.2% | Profitable, slow |
| M06 (not in the item list) | Not in the item list | 120,120 | 3,669 | n/a | 2.4% | Cost unknown |
| Hummus | Sides | 119,209 | 8,582 | 71.2% | 2.4% | Profitable, slow |
| Soft Drink | Drinks | 82,583 | 11,894 | 71.2% | 1.7% | Profitable, slow |
| Water | Drinks | 47,478 | 11,966 | 74.8% | 1.0% | Profitable, slow |

Source: tables/items.csv

## 5. Guests

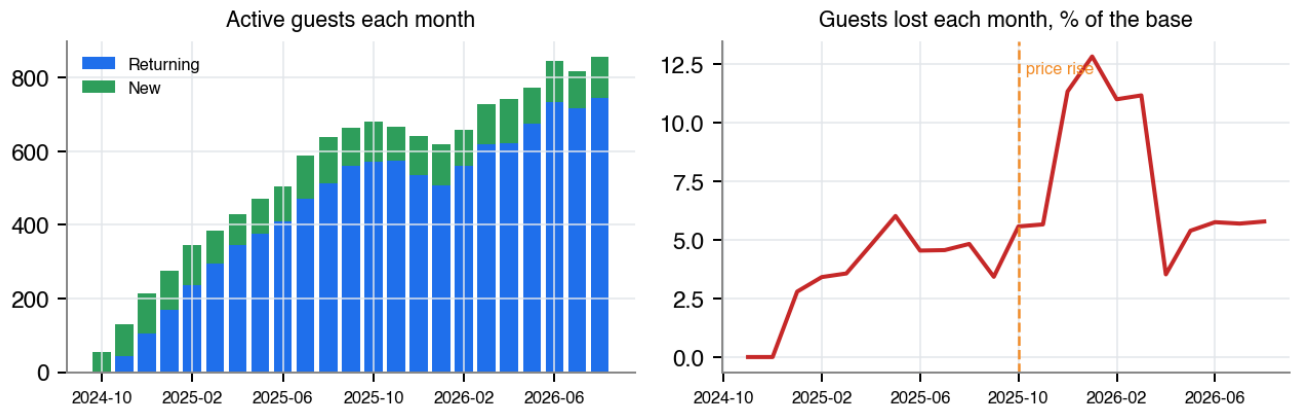



Figure 5. New and returning guests, and the share lost each month. Source: tables/customers_monthly.csv

Known guests account for 28% of sales; the rest pay without being identified. Of new ones, 89% come back within 90 days, and a new guest brings SAR 1,115 of sales in the first year.

| Group | Guests | Sales | Average visits | Days since last (median) | Share of sales |
|---|---|---|---|---|---|
| At risk | 36 | 86,878 | 32.8 | 68 | 4% |
| Lost | 1219 | 909,901 | 10.4 | 321 | 37% |
| Loyal | 372 | 1,014,448 | 37.0 | 6 | 41% |
| New | 230 | 51,134 | 3.0 | 8 | 2% |
| Occasional | 610 | 393,720 | 8.7 | 21 | 16% |

Source: tables/segments.csv

## 6. How guests reacted to prices

| Category | Month | Price | Units | Elasticity | Lost a month, before | Lost a month, after |
|---|---|---|---|---|---|---|
| Mains | 2025-10 | +15.9% | -6.9% | -0.43 | 4.3% | 8.8% |

Source: tables/price_events.csv

Elasticity is the change in units for each 1% of price: -0.5 means a 10% rise loses 5% of units. It is measured over the three months either side, against all orders, so it mixes the price with whatever else changed at the time.

## 7. Marketing

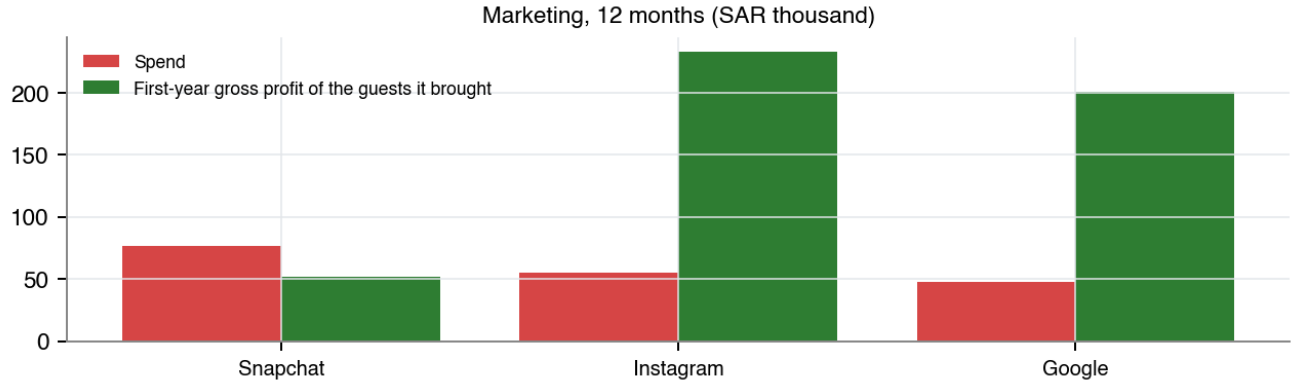



Figure 6. Spend against the first-year gross profit of the guests each channel brought. Source: tables/marketing.csv

| Channel | Spend | Clicks | New guests | Cost of each | First-year gross profit | Return per unit |
|---|---|---|---|---|---|---|
| Snapchat | 76,564 | 125,821 | 80 | 957 | 51,711 | 0.68 |
| Instagram | 54,731 | 89,731 | 360 | 152 | 232,699 | 4.25 |
| Google | 47,621 | 76,382 | 310 | 154 | 200,379 | 4.21 |

Source: tables/marketing.csv

A return below one means the channel cost more than its guests earned back in their first year. Guests who joined through no channel are not counted, so the returns are a floor.

## 8. People, hours and busy times

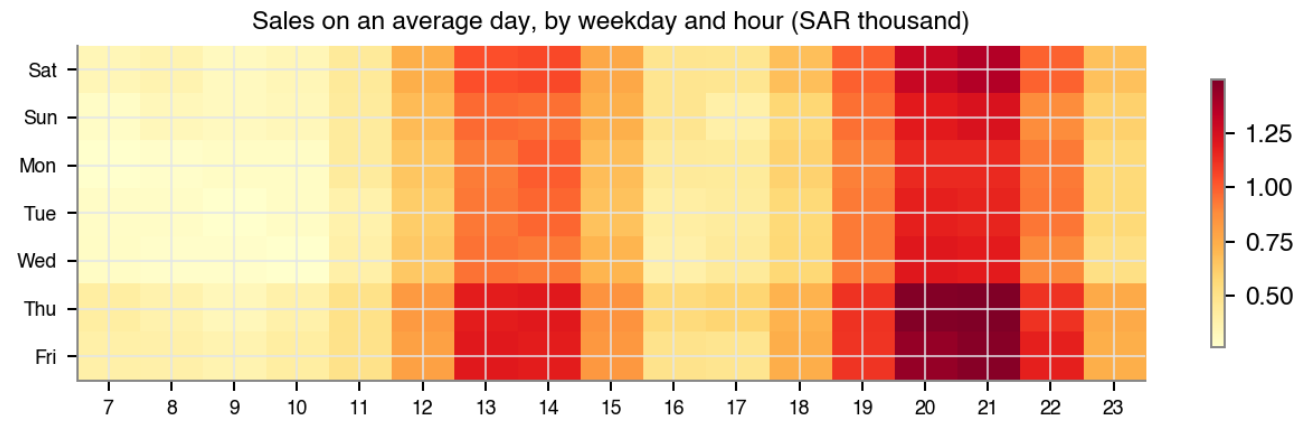



Figure 7. When sales happen. Source: tables/busy.csv

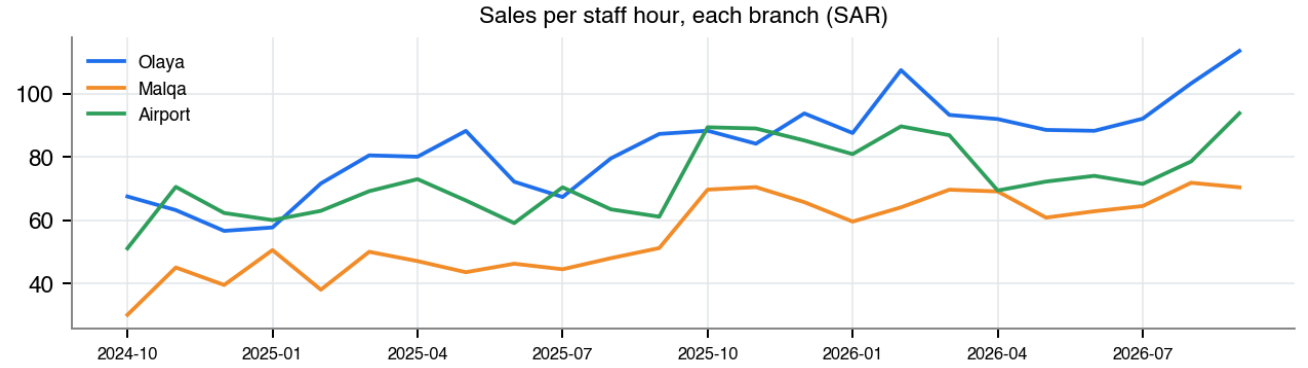



Figure 8. Sales per staff hour by branch. Source: tables/place_month.csv

The measures restaurants and cafés usually watch:

| Measure | Airport | All | Malqa | Olaya |
|---|---|---|---|---|
| Average order value | 74 | 74 | 74 | 75 |
| Food cost | 39.2% | 39.4% | 39.3% | 39.7% |
| Orders per seat a day | 1.33 | 0.97 | 0.75 | 0.90 |
| Sales per labour hour | 81 | 81 | 66 | 94 |
| Sales per square metre a month | 831 | 593 | 461 | 546 |

Source: tables/kpis.csv

## 9. Against target

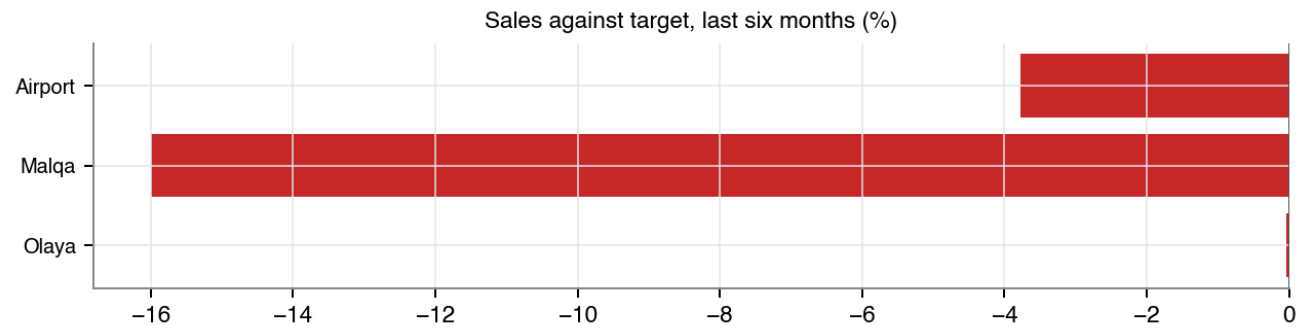



Figure 9. Sales against target over the last six months. Source: tables/targets.csv

| Month | Branch | Measure | Target | Actual | Gap |
|---|---|---|---|---|---|
| 2026-07 | Olaya | revenue | 165,900 | 166,977 | +0.6% |
| 2026-07 | Malqa | revenue | 123,200 | 102,448 | -16.8% |
| 2026-07 | Airport | revenue | 151,900 | 144,482 | -4.9% |
| 2026-08 | Olaya | revenue | 181,100 | 177,859 | -1.8% |
| 2026-08 | Malqa | revenue | 122,000 | 103,703 | -15.0% |
| 2026-08 | Airport | revenue | 152,900 | 146,105 | -4.4% |
| 2026-09 | Olaya | revenue | 171,900 | 173,434 | +0.9% |
| 2026-09 | Malqa | revenue | 120,700 | 102,264 | -15.3% |
| 2026-09 | Airport | revenue | 155,500 | 151,038 | -2.9% |
| 2026-07 | Olaya | orders | 2,134 | 2,200 | +3.1% |
| 2026-07 | Malqa | orders | 1,579 | 1,362 | -13.7% |
| 2026-07 | Airport | orders | 2,015 | 1,976 | -1.9% |
| 2026-08 | Olaya | orders | 2,289 | 2,360 | +3.1% |
| 2026-08 | Malqa | orders | 1,620 | 1,397 | -13.8% |
| 2026-08 | Airport | orders | 2,019 | 1,980 | -1.9% |
| 2026-09 | Olaya | orders | 2,220 | 2,289 | +3.1% |
| 2026-09 | Malqa | orders | 1,591 | 1,372 | -13.8% |
| 2026-09 | Airport | orders | 2,066 | 2,026 | -1.9% |

Source: tables/targets.csv

## 10. What a new branch would earn

Sized like the 3 existing branches: SAR 123k of sales a month once mature, a gross margin of 60.6%, rent of SAR 18k and staff of SAR 30k a month. That leaves SAR 7,755 a month when mature, paying back the fit-out in about 58 months. The existing branches range from SAR 1,692 to 2,992 of sales per seat a month, so location decides most of the answer. The dashboard lets the owner change every assumption.

## 11. The data received

Every line exported was read; what could not be used is listed here with the reason. The data readiness note says what to collect next, and how.

| Source | Table | Field | What happened | Rows | Example |
|---|---|---|---|---|---|
| pos | transactions | | rows read / kept | 257,034 | kept 251732 |
| pos | transactions | status | excluded by rule: ['void'] | 5,302 | |
| menu | items | | rows read / kept | 15 | kept 15 |
| branches | places | | rows read / kept | 3 | kept 3 |
| ledger | expenses | | rows read / kept | 480 | kept 480 |
| payroll | staff | | rows read / kept | 72 | kept 72 |
| ads | marketing | | rows read / kept | 2,190 | kept 2190 |
| loyalty | parties | | rows read / kept | 2,730 | kept 2730 |
| budget | targets | | rows read / kept | 144 | kept 144 |
| | transactions | item_id | not found in items | 2,906 | M06 |
| | expenses | place_id | not found in places | 48 | HQ |

Source: tables/intake_log.csv

Days on which a branch that trades every day has no data:

| place_id | date | issue |
|---|---|---|
| B1 | 2026-02-14 | no transactions on a day the site normally trades |

Source: tables/data_gaps.csv

## 12. All health tests

| Area | Test | Result | Threshold | Status |
|---|---|---|---|---|
| Data | Share of exported lines that could be used | 97.9% | pass >= 97.0%; fail < 90.0% | PASS |
| Data | Days since the latest transaction | 5 | pass <= 10; fail > 35 | PASS |
| Data | Trading days with no data | 1 | pass <= 0; fail > 3 | WARN |
| Data | Sales whose cost is unknown | 1.4% | pass <= 1.0%; fail > 5.0% | WARN |
| Profit | Operating margin after head office, last 12 months | -7.9% | pass >= 5.0%; fail < 0.0% | FAIL |
| Profit | Share of sales made at loss-making sites | 36.1% | pass <= 0.0%; fail > 25.0% | FAIL |
| Profit | Gross margin, last 12 months | 60.6% | pass >= 60.0%; fail < 50.0% | PASS |
| Items | Share of sales from items with a low margin | 16.0% | pass <= 5.0%; fail > 15.0% | FAIL |
| Sales | Sales growth, recent months | 23.7% (the same months a year earlier) | pass >= 0.0%; fail < -5.0% | PASS |
| Customers | New customers who come back within 90 days | 89.1% | pass >= 40.0%; fail < 25.0% | PASS |
| Customers | Customers lost each month, change after the latest price rise | 107.2% (Mains, 2025-10) | pass <= 20.0%; fail > 50.0% | FAIL |
| Customers | Units lost for each 1% of price rise | 0.43 (Mains, 2025-10) | pass <= 0.50; fail > 1.00 | PASS |
| Marketing | Worst channel: first-year gross profit per unit of spend | 0.68 (Snapchat) | pass >= 1.00; fail < 0.50 | WARN |
| Operations | Weakest site: sales per staff hour against the best | 70.7% | pass >= 80.0%; fail < 60.0% | WARN |
| Planning | Site-months more than 5% below target, last 6 months | 33.3% | pass <= 20.0%; fail > 50.0% | WARN |
| Planning | Forecast error on months held back | 10.9% | pass <= 10.0%; fail > 20.0% | WARN |
| Sales | Share of sales from the largest site | 39.5% | pass <= 50.0%; fail > 70.0% | PASS |

Source: tables/tests.csv

# Limitations and assumptions

- The analysis shows what happened and what moved together. It does not prove cause: a price change and a fall in guests may share another reason. A controlled test proves cause.
- Profit by branch is only as good as the way costs are booked to branches in the accounts.
- Guests who pay without being identified are outside the guest analysis.
- The forecast extends the past. It cannot see a new competitor, a lost contract or a change in the market.

# Appendix

## Preparation log

| step | action | rows_affected |
|---|---|---|
| transactions | identical export lines removed | 8121 |
| transactions | after the period end 2026-09-30 left out | 0 |
| transactions | refunds kept as negative sales | 512 |
| transactions | cost taken from the item list (quantity x unit cost) | 240774 |
| transactions | no cost known (item missing from the item list) | 2837 |
| transactions | trading days with no data (listed) | 1 |

Source: tables/cleaning_log.csv

## Data quality

### expenses · 480 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

### items · 15 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

### marketing · 2190 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

### parties · 2730 rows · PASSED (0 errors, 1 warnings)

| Check | Severity | Column | Count | Message |
|---|---|---|---|---|
| missing_values | warning | segment | 2730 | 2730 missing (100.00%) |

Source: data_quality/dq_parties.json

### places · 3 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

### staff · 72 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

### targets · 144 rows · PASSED (0 errors, 0 warnings)

No data-quality issues detected.

### transactions · 251732 rows · PASSED (0 errors, 8 warnings)

| Check | Severity | Column | Count | Message |
|---|---|---|---|---|
| missing_values | warning | line_no | 251732 | 251732 missing (100.00%) |
| missing_values | warning | party_id | 181522 | 181522 missing (72.11%) |
| missing_values | warning | unit_price | 251732 | 251732 missing (100.00%) |
| missing_values | warning | discount | 251732 | 251732 missing (100.00%) |
| missing_values | warning | cost | 251732 | 251732 missing (100.00%) |
| missing_values | warning | tax | 251732 | 251732 missing (100.00%) |
| missing_values | warning | staff_id | 251732 | 251732 missing (100.00%) |
| duplicate_rows | warning | | 8121 | 8121 fully duplicated rows |

Source: data_quality/dq_transactions.json

### transactions_ready · 243611 rows · PASSED (0 errors, 1 warnings)

| Check | Severity | Column | Count | Message |
|---|---|---|---|---|
| missing_values | warning | cost | 2837 | 2837 missing (1.16%) |

Source: data_quality/dq_transactions_ready.json

## Audit trail

| | |
|---|---|
| Run ID | `ba_20261111T142253Z_f9025f` |
| Run started / finished (UTC) | 2026-11-11T14:22:53+00:00 |
| Analyst | info@proquantanalytics.com |
| Code version (git) | `3a6bb15275c5 (uncommitted changes) · core 0.1.0` |
| Config SHA-256 | `7143263c7fcb21d5` |
| Input: loyalty | `2730 rows · sha256 5cdd91850ba23aa6` |
| Input: ads | `2190 rows · sha256 2141427621ddf485` |
| Input: budget | `72 rows · sha256 92fb307eeb3613a0` |
| Input: pos | `257034 rows · sha256 7d843af5e446cb27` |
| Input: ledger | `480 rows · sha256 5366f65dc6e5365a` |
| Input: branches | `3 rows · sha256 7213bda7ecaa7296` |
| Input: payroll | `72 rows · sha256 9ce7401c654058f1` |
| Input: menu | `15 rows · sha256 d9013d82a0d97fe6` |
| Environment | `python 3.13.5 · pandas 2.2.3 · numpy 2.1.3` |

Source: audit.json in the run folder

This report was prepared by ProQuant Analytics for the exclusive use of Demo Client under the terms of the engagement agreement. It is based on data provided by the client and on the methodology and assumptions described herein, which the client should review. Past or simulated performance is not a reliable indicator of future results. This report does not constitute investment, legal, tax or accounting advice.